University of Novi Sad
Faculty of Technical Sciences
Chair of Applied Computer Science
Data Science and Information Systems Group

Rapid Tool Prototyping

Eclipse Sirius Framework for the Development of Graphical DSLs

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DAIS
Data Science and Information Systems Group

Agenda

- DSL Development
- Rapid Tool Prototyping
- Feature Oriented Domain Analysis
- Abstract Syntax
- Concrete Syntax (Frameworks)
- Evaluation
- Sirius – Additional

Domain-Specific Languages

- **Model-Driven paradigm**
  - Lead system development
- **Users focused on domain issues**
- **Formal language**
  - Machine-readable models
- **Modeling tools**

DSL Development Process

- **Requirements**
  - Given or researched
- **Feature Oriented Domain Analysis (FODA)**
  - Domain Concepts
- **Abstract Syntax**
  - Meta-Model
- **Concrete Syntax**
  - Textual / Graphical
- **Evaluation**
  - FQAD

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Rapid Tool Development – when to develop?

- **Prototype tool**
  - Test a solution
- **Research projects**
  - Changeable or undefined requirements
- **Software product?**

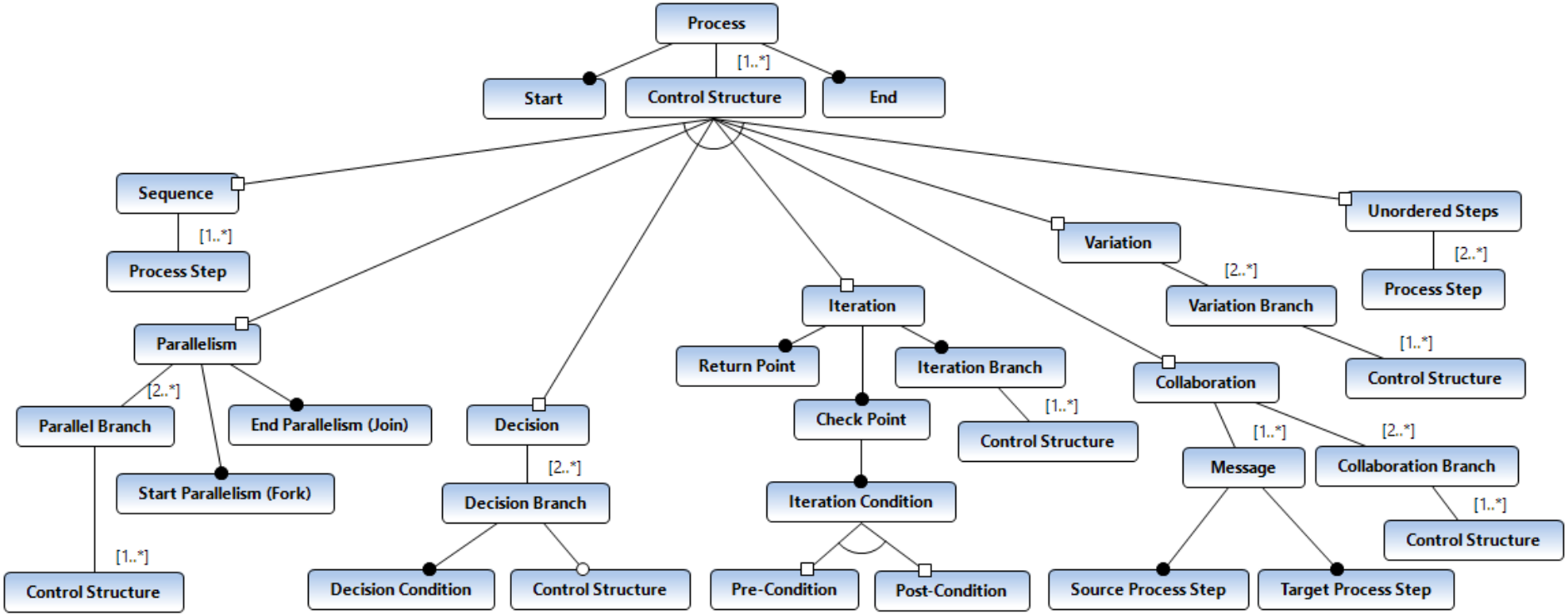
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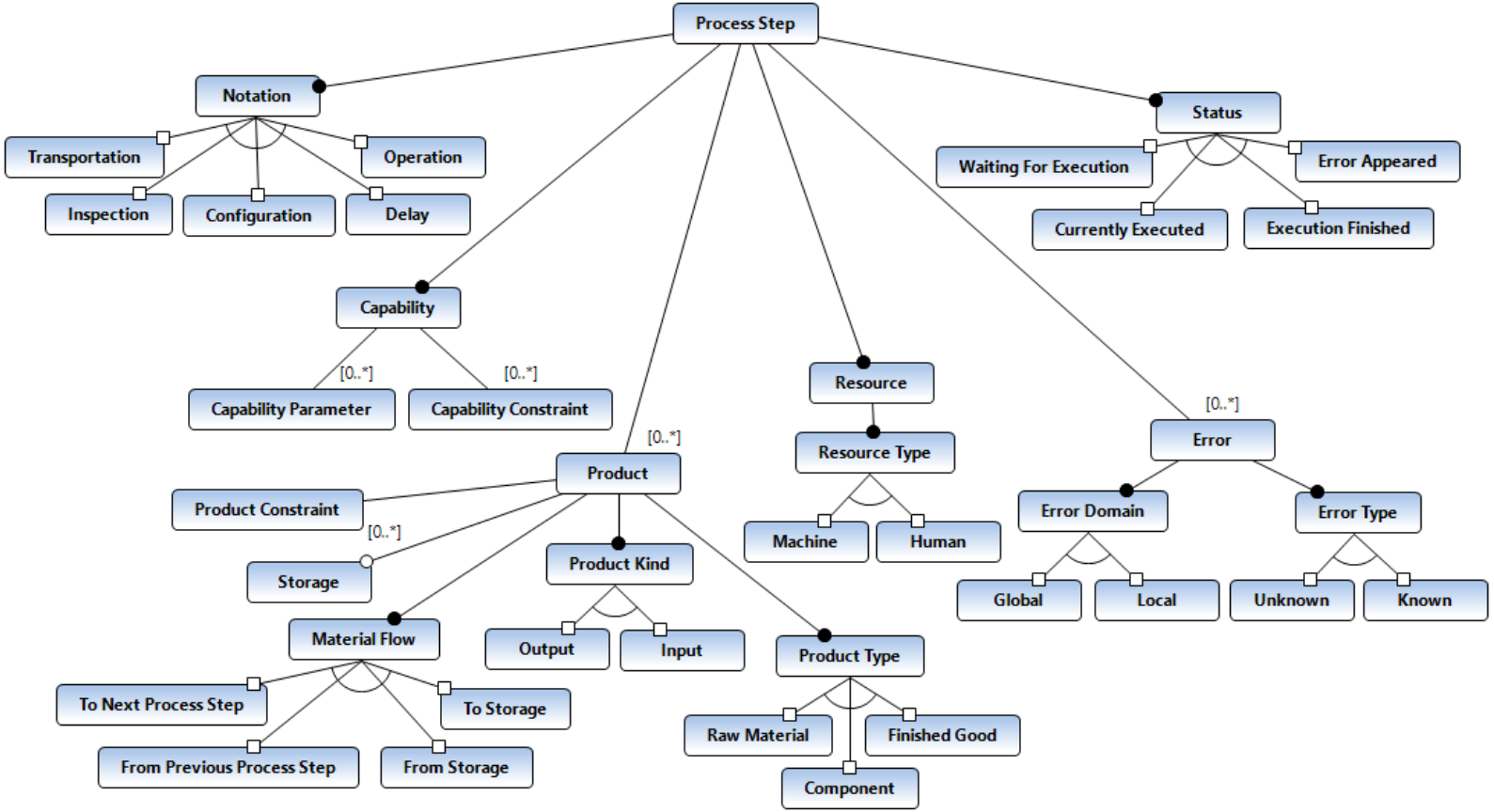
Feature Oriented Domain Analysis (FODA)

- **FODA method** (Kang et al. 1990)
  - **Capture domain experts' knowledge**
  - **Analyze domain and identify features** usually expected to support software development
  - Features to be described in a **generic way** – widely applicable
- **Feature**
  - "any prominent and distinctive **aspect or characteristic that is visible to various stakeholders** (i.e., end-users, domain experts, developers, etc.)"
- **Cardinality-based FODA**

MultiProLan – FODA



MultiProLan – FODA



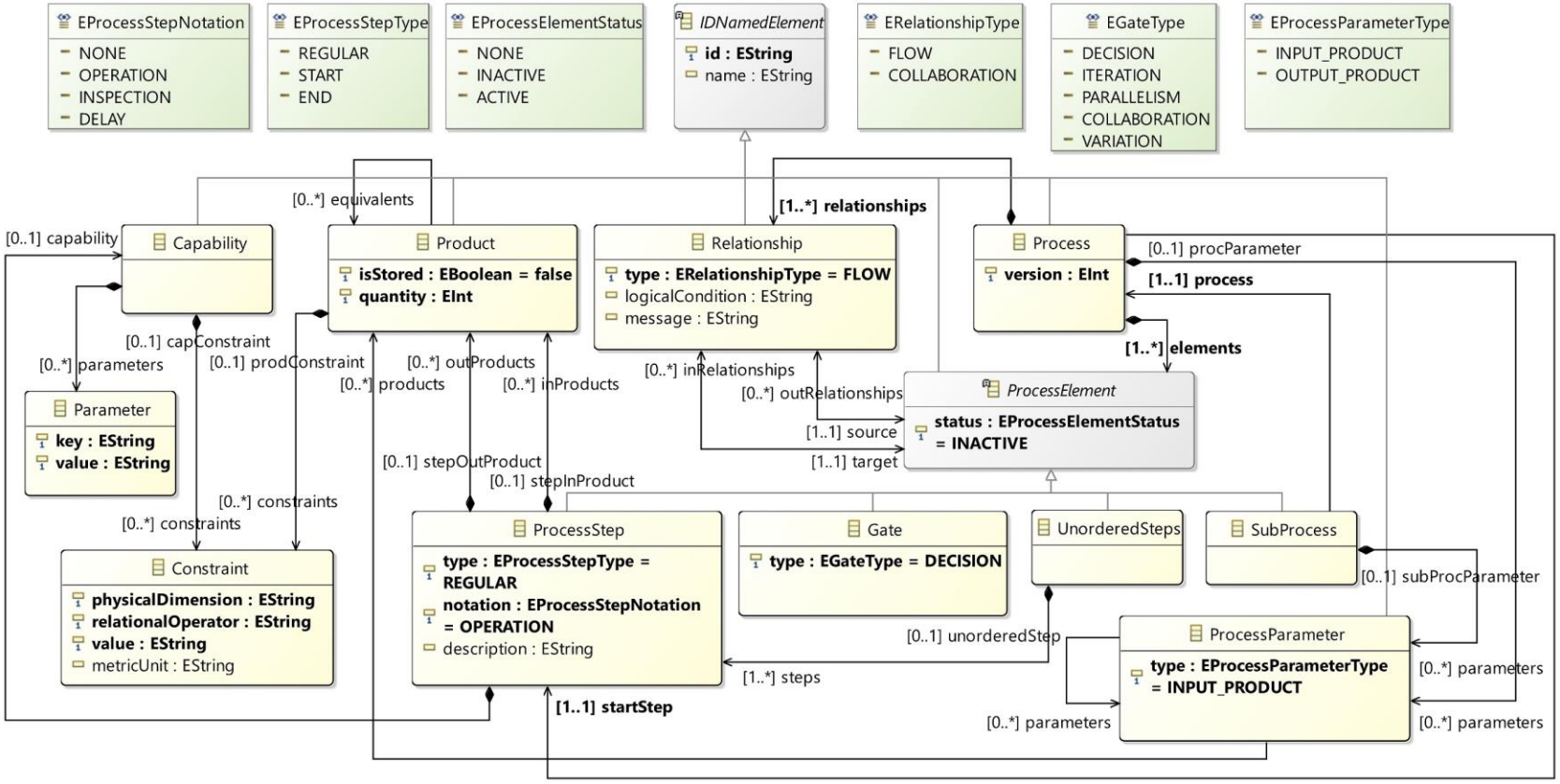
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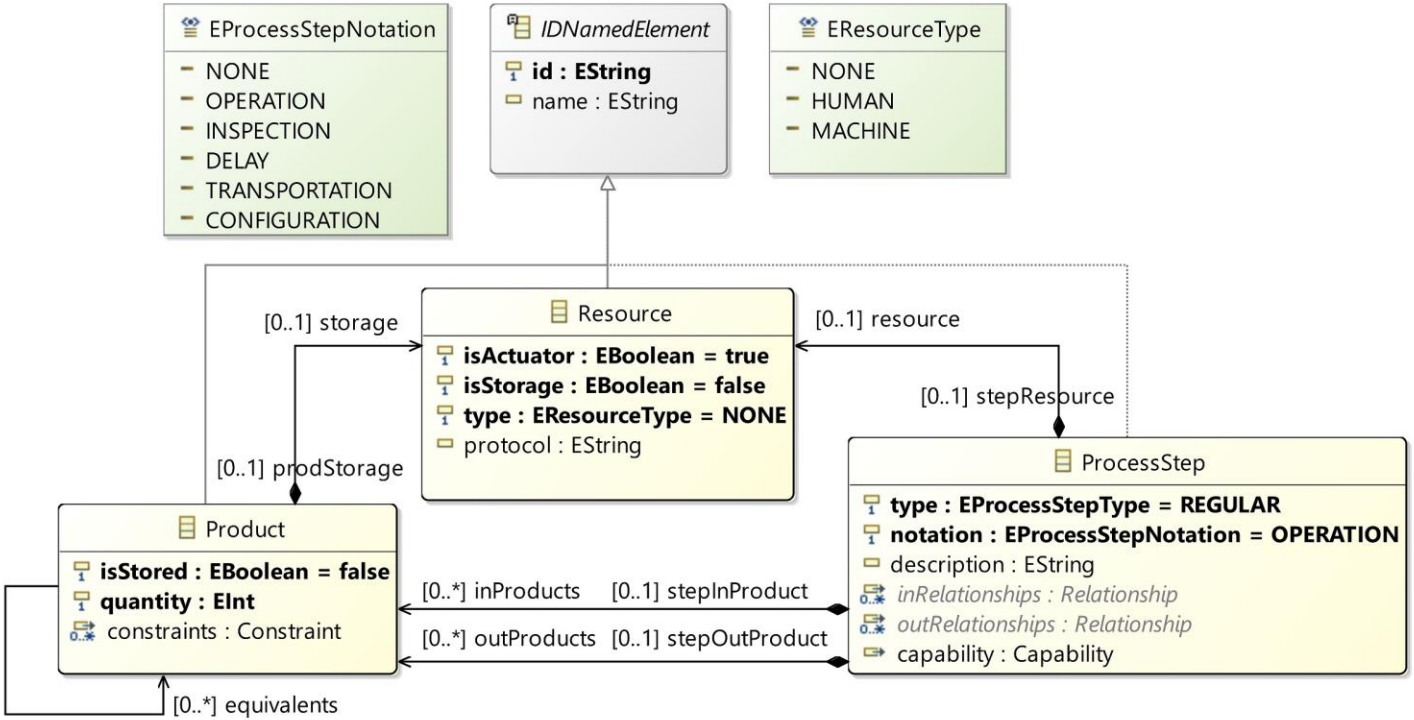
Abstract Syntax

- **Meta-model**
- **Domain concepts, their properties and relationships**
  - Constraints are only partially described
- **Constraints**
  - Specified using a constraints language
  - Object Constraint Language (OCL)

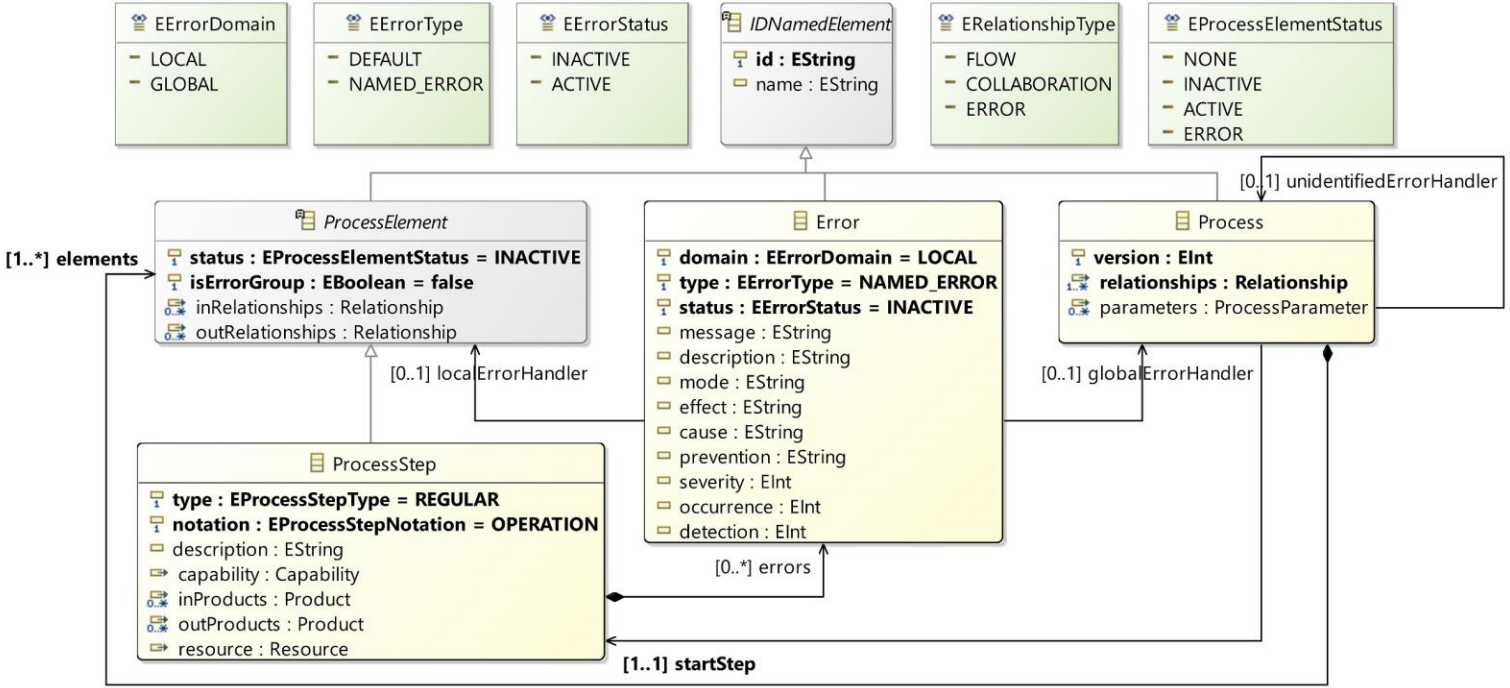
MultiProLan – Abstract Syntax



MultiProLan – Abstract Syntax



MultiProLan – Abstract Syntax



MultiProLan – OCL Examples

formalno, X je ključ ako

- $1^0 \quad (\forall u, v \in r(R))(u[X] = v[X] \Rightarrow u = v)$
- $2^0 \quad (\forall Y \subset X)(\neg 1^0)$

- Ne sme da postoji više od jednog početnog procesnog koraka:

```
invariant oneStartProcessStep:
  ProcessStep.allInstances()->forAll(ps1, ps2 |
    (ps1.type = EProcessStepType::START and ps2.type = EProcessStepType::START)
      implies (ps1 = ps2));
```

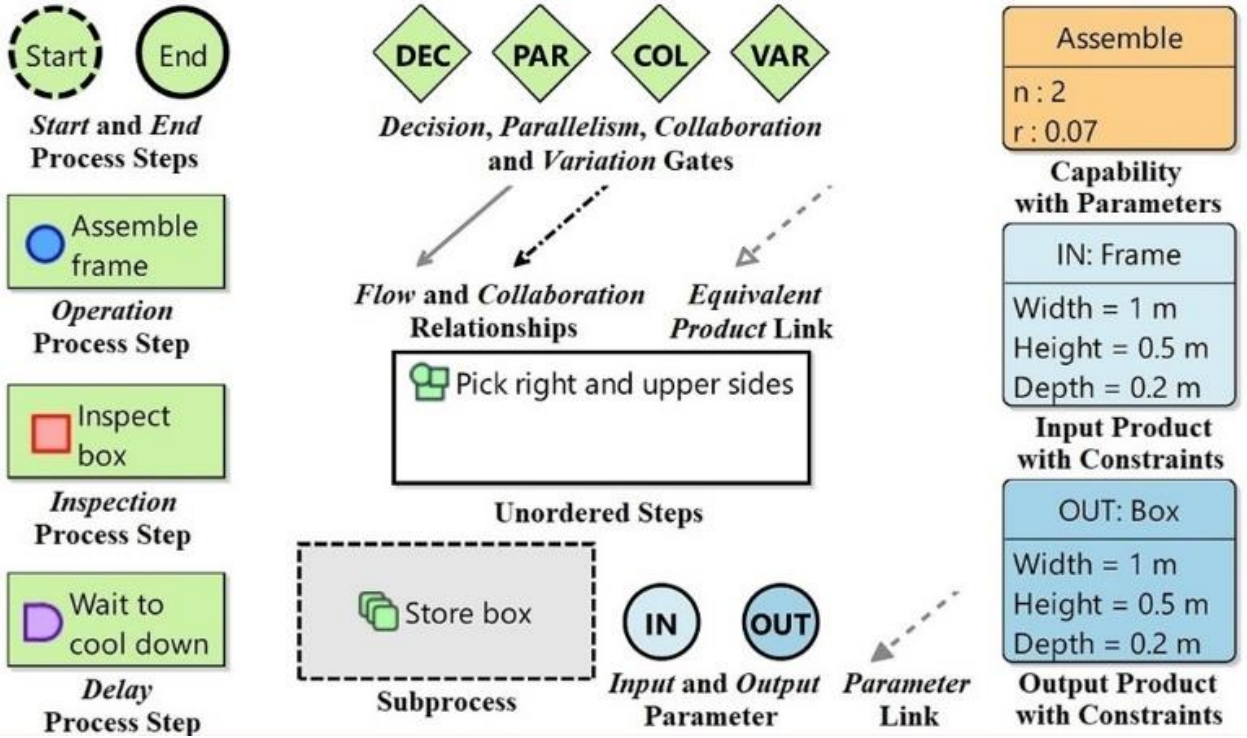
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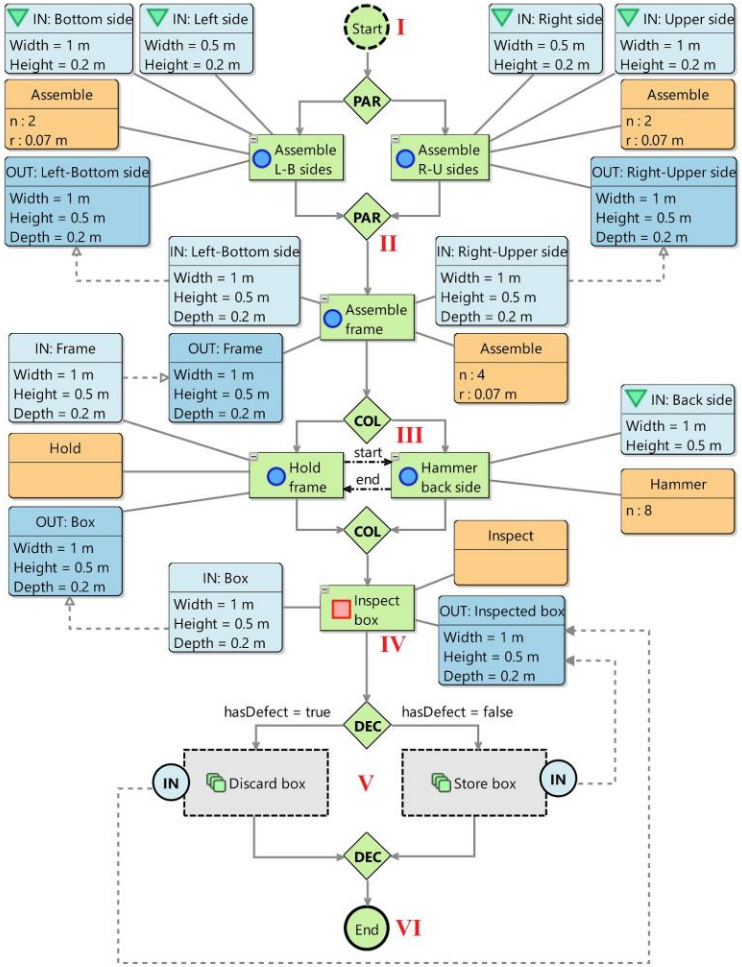
Concrete Graphical Syntax

- **Advantages**
  - Model structure is easier to comprehend
  - Easier model navigation (e.g., zooming)
  - Visual languages are easier to learn
- **Disadvantages**
  - Harder to develop and maintain
  - Serialization format is different from the presentation format
  - Diagrams may become enormous (e.g., sub-elements, filters, layers)

MultiProLan – Concrete Syntax



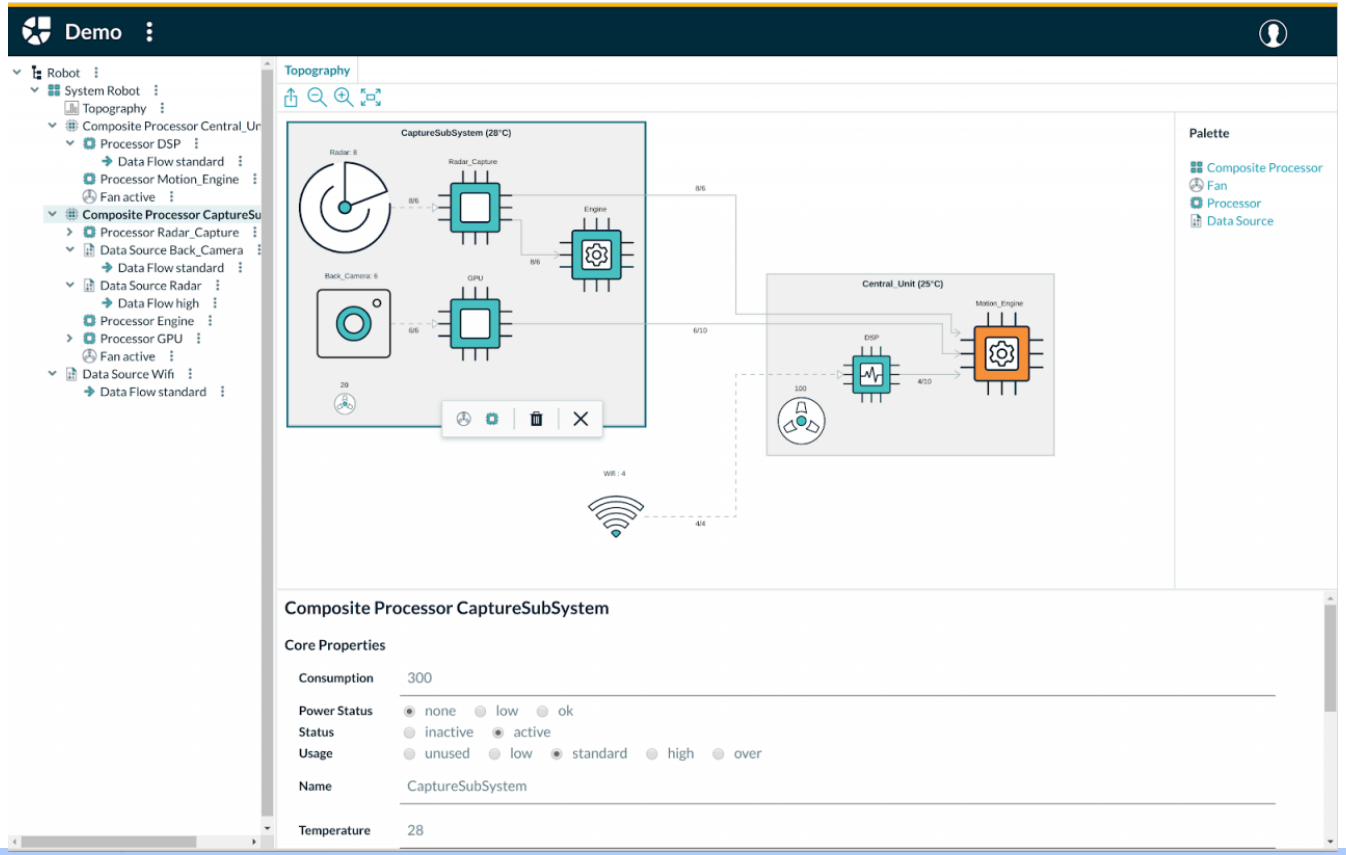
MultiProLan – Example



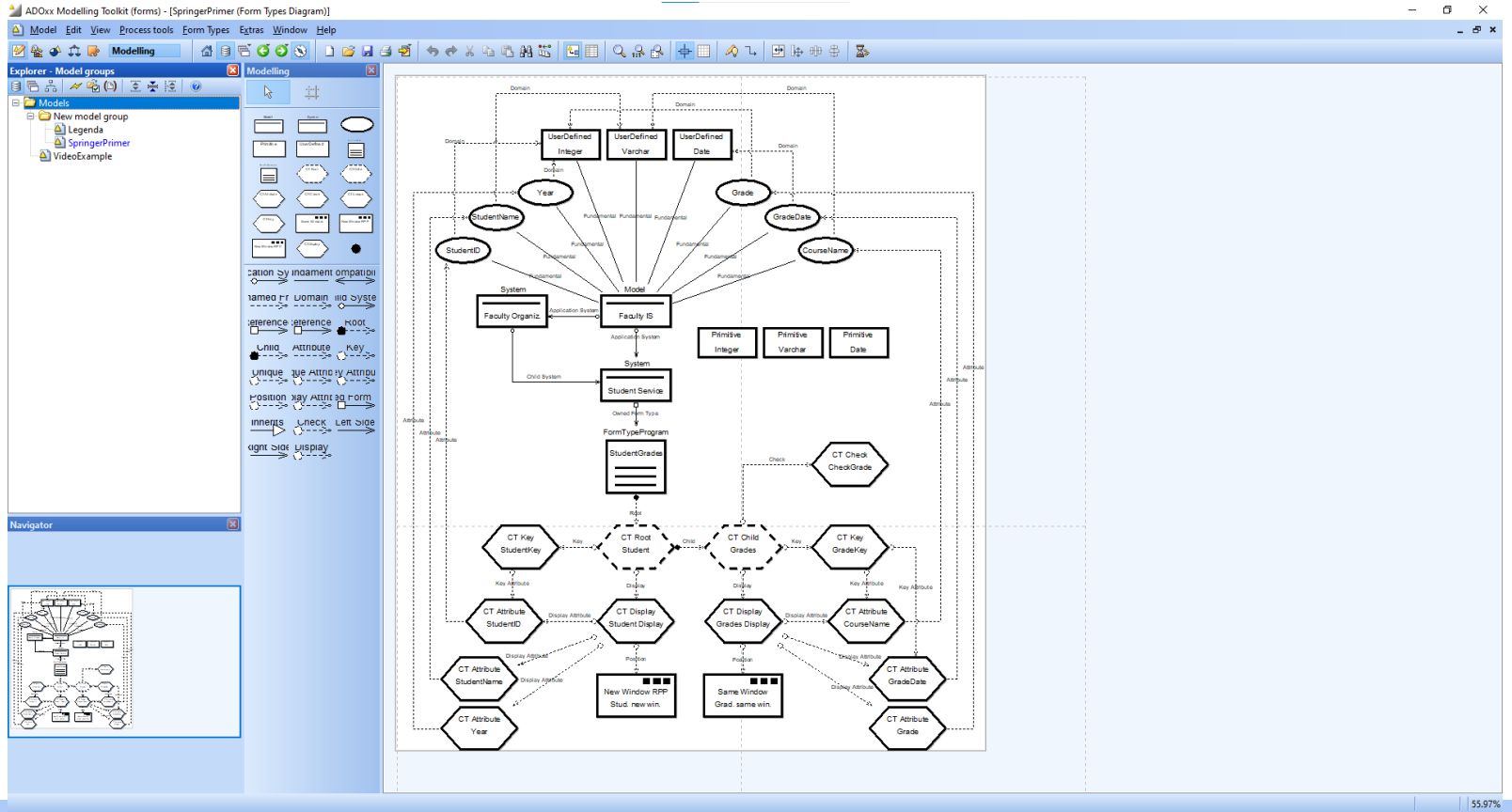
Concrete Graphical Syntax Frameworks

- **Sirius** (EMF, Obeo)
- **Sirius Web** (Obeo)
- **ADOxx** (OMiLAB)

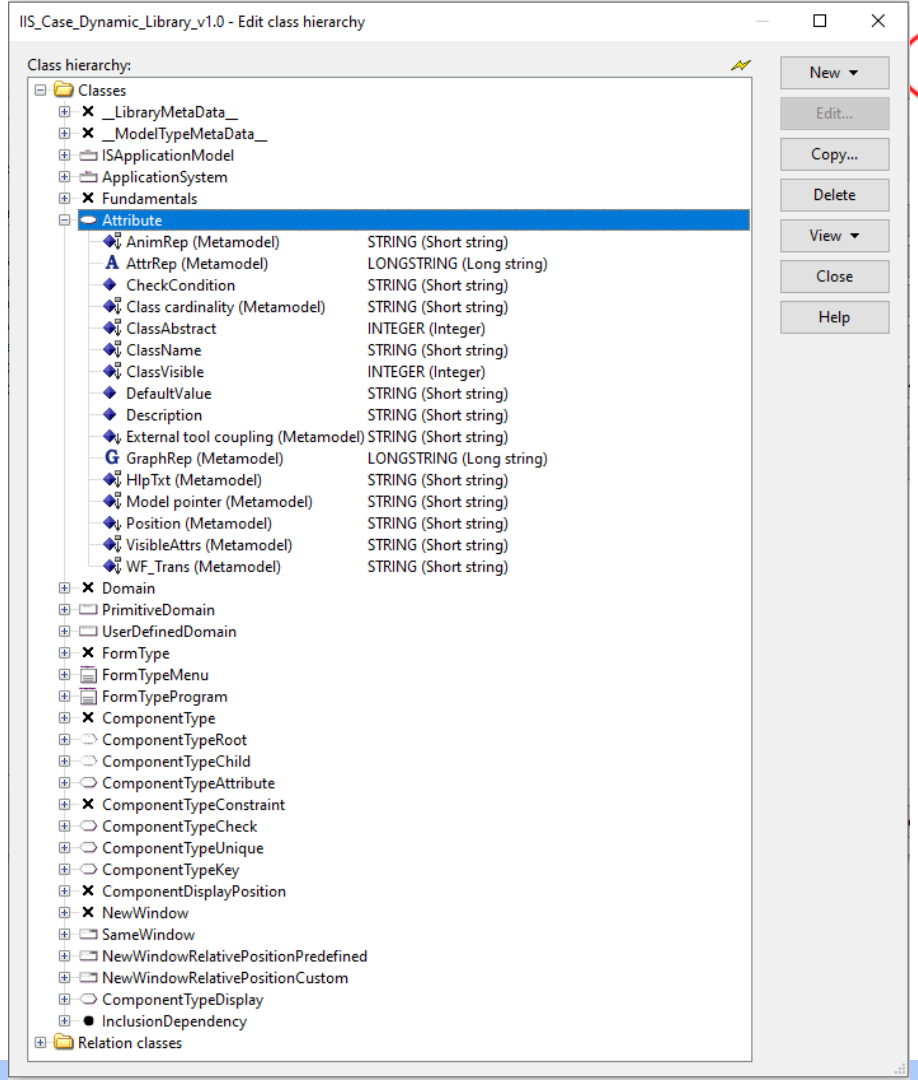
Obeo Sirius Web



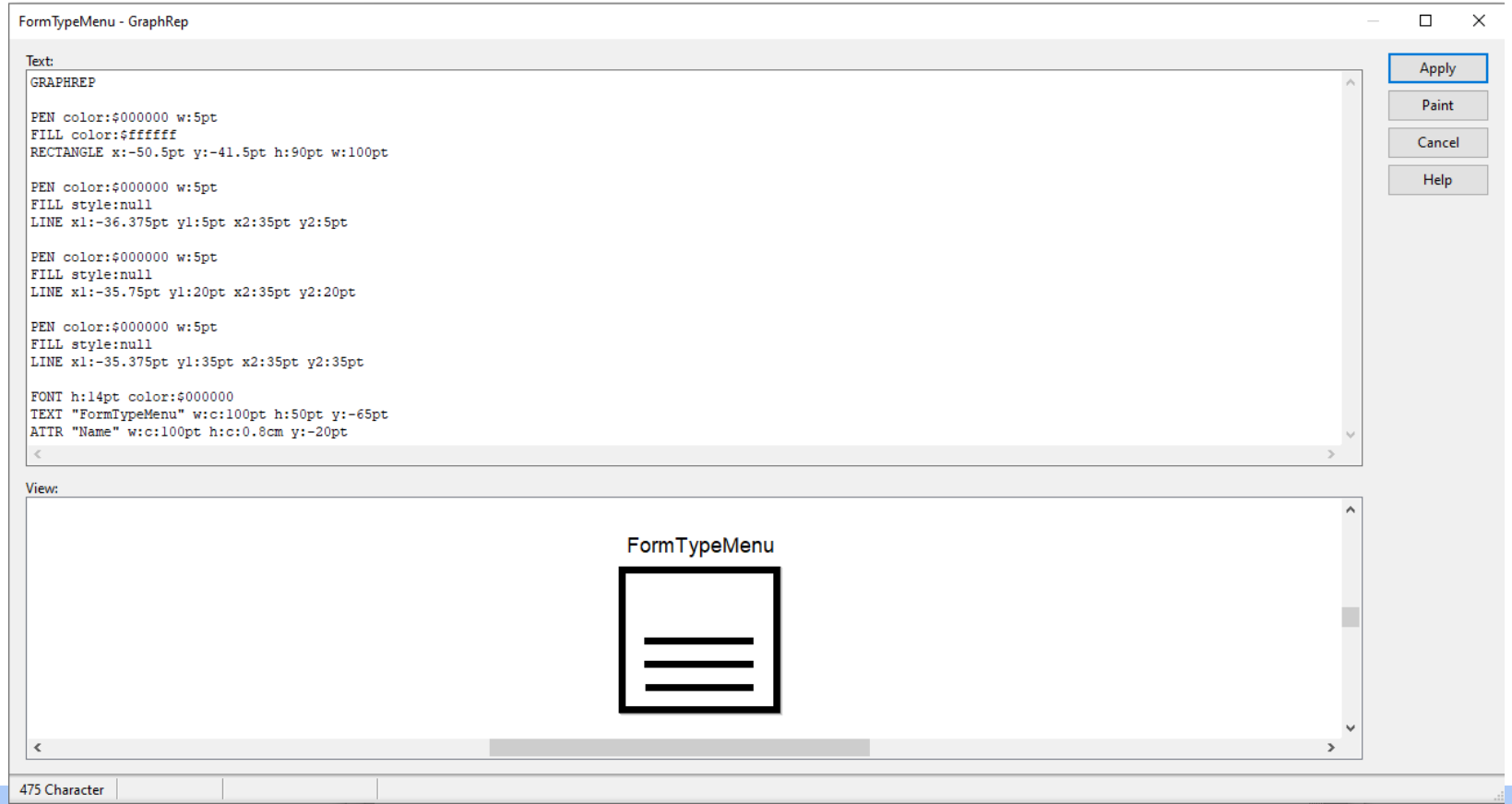
ADOxx – Tool Example



ADOxx – Abstract Syntax



ADOxx – Concrete Syntax



ADOxx – AdoScript

```
#Get the active model
CC "Modeling" GET_ACT_MODEL
SET nCurrentModelId:(modelid)
    … … …
#Create an SQL file
CC "AdoScript" FILE_DIALOG saveas filter1:"SQL Files" type1:"*.sql" default-ext:"sql"
SET sSQLFileLocation:(path)

#Get ComponentTypeRoot and ComponentTypeChild objects
CC "Core" GET_ALL_OBJS_OF_CLASSNAME modelid:(nCurrentModelId) classname:"ComponentTypeRoot"
SET sComponentTypeIds:(objids)
CC "Core" GET_ALL_OBJS_OF_CLASSNAME modelid:(nCurrentModelId) classname:"ComponentTypeChild"
SET sComponentTypeIds:(sComponentTypeIds + " " + objids)
```

```
CREATE TABLE Student (
    StudentID Integer NOT NULL,
    StudentName Varchar(30) NOT NULL,
    Year Integer NOT NULL
);

CREATE TABLE Grades (
    CourseName Varchar(30) NOT NULL,
    GradeDate Date NOT NULL,
    Grade Integer DEFAULT 5 NOT NULL,
    StudentID Integer
);

ALTER TABLE Student ADD CONSTRAINT StudentKey PRIMARY KEY(StudentID);

ALTER TABLE Grades ADD CONSTRAINT GradeKey PRIMARY KEY(CourseName, GradeDate, StudentID);
ALTER TABLE Grades ADD CONSTRAINT CheckGrade CHECK(Grade >= 5 AND Grade <= 10);
ALTER TABLE Grades ADD CONSTRAINT Grades_Student_FK FOREIGN KEY(StudentID) REFERENCES Student(StudentID);
```

```
        #Create alter table statement and write it into the file
        SET sTextValue:("ALTER TABLE " + sCTName + " ADD CONSTRAINT " + sCheckName + " CHECK(" + sCheckCondition + ");\n")
        CC "AdoScript" FWRITE file:(sSQLFileLocation) text:(sTextValue) append:1
      }
    }
    … … …
}
```

ADOxx

- **Advantages**
  - ADOxx textual syntax for specifying meta-model
    - GraphRep Generator Tool
  - Single framework only
  - Less framework configurations (tool pallet, menu, object property section)
  - User management system
  - Deployment
- **Disadvantages**
  - ADOxx textual syntax for specifying meta-model
    - No class diagram
    - Difficult for beginners
  - AdoScript

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Evaluation

- **Framework for Qualitative Assessment of Domain-specific languages (FQAD)** (Kahraman and Bilgen 2015)
  - To **evaluate** a language and a modeling tool based on quality characteristics
  - Evaluate the language and the tool based on their **functional suitability, usability, reliability, expressiveness, and productivity characteristics**
  - **…**

Experience					
Question	**Inexperienced**	**Relatively inexperienced**	**Medium experienced**	**Relatively experienced**	**Experienced**
How would you describe your previous experience in designing business processes?	20%	12%	40%	20%	8%
How would you describe your previous experience in designing production processes?	56%	20%	16%	4%	4%
How would you describe your previous experience with Computer Aided Software Engineering (CASE) tools for modeling?	8%	20%	28%	28%	16%
Functional suitability					
Question	**Very low**	**Low**	**Medium**	**High**	**Very high**
How would you describe the scope of production process domain concepts and scenarios that can be expressed in MultiProLan?	0%	0%	0%	64%	36%
How would you describe MultiProLan's level of suitability for the production process specification?	0%	0%	4%	64%	32%
Usability					
Question	**Strongly disagree**	**Disagree**	**No opinion**	**Agree**	**Strongly agree**
MultiProLan language elements are understandable.	0%	4%	0%	36%	60%
The concepts and symbols of MultiProLan are learnable and rememberable.	0%	0%	0%	20%	80%
MultiProLan has capability to help users achieve their tasks in an acceptable number of steps.	0%	0%	8%	52%	40%
MultiProLan is appropriate for your needs.	0%	0%	48%	16%	36%
MultiProLan Eclipse environment has elements that facilitate to operate and control the language.	0%	0%	20%	52%	28%
MultiProLan has graphical symbols that are good looking/attractive.	0%	8%	0%	44%	48%
By separating MasL and DetL models, users can model production processes easier.	0%	0%	4%	32%	64%
By creating different modeling layers, models become more readable.	0%	0%	0%	12%	88%
Reliability					
Question	**Strongly disagree**	**Disagree**	**No opinion**	**Agree**	**Strongly agree**
MultiProLan protects users against making errors.	0%	0%	16%	60%	24%
MultiProLan has a functional model validator.	0%	4%	28%	32%	36%
Expressiveness					
Question	**Strongly disagree**	**Disagree**	**No opinion**	**Agree**	**Strongly agree**
A problem-solving strategy can be mapped into a specification easily.	0%	0%	0%	64%	36%
MultiProLan is at the right abstraction level such that it is not more complex or detailed than necessary.	0%	0%	0%	52%	48%
MultiProLan provides one and only one good way to express every concept of interest.	0%	12%	20%	48%	20%
Productivity					
Question	**Long**	**Relatively long**	**Medium**	**Relatively short**	**Short**
How would you describe the specification time of a production process model with MultiProLan?	0%	4%	28%	52%	16%

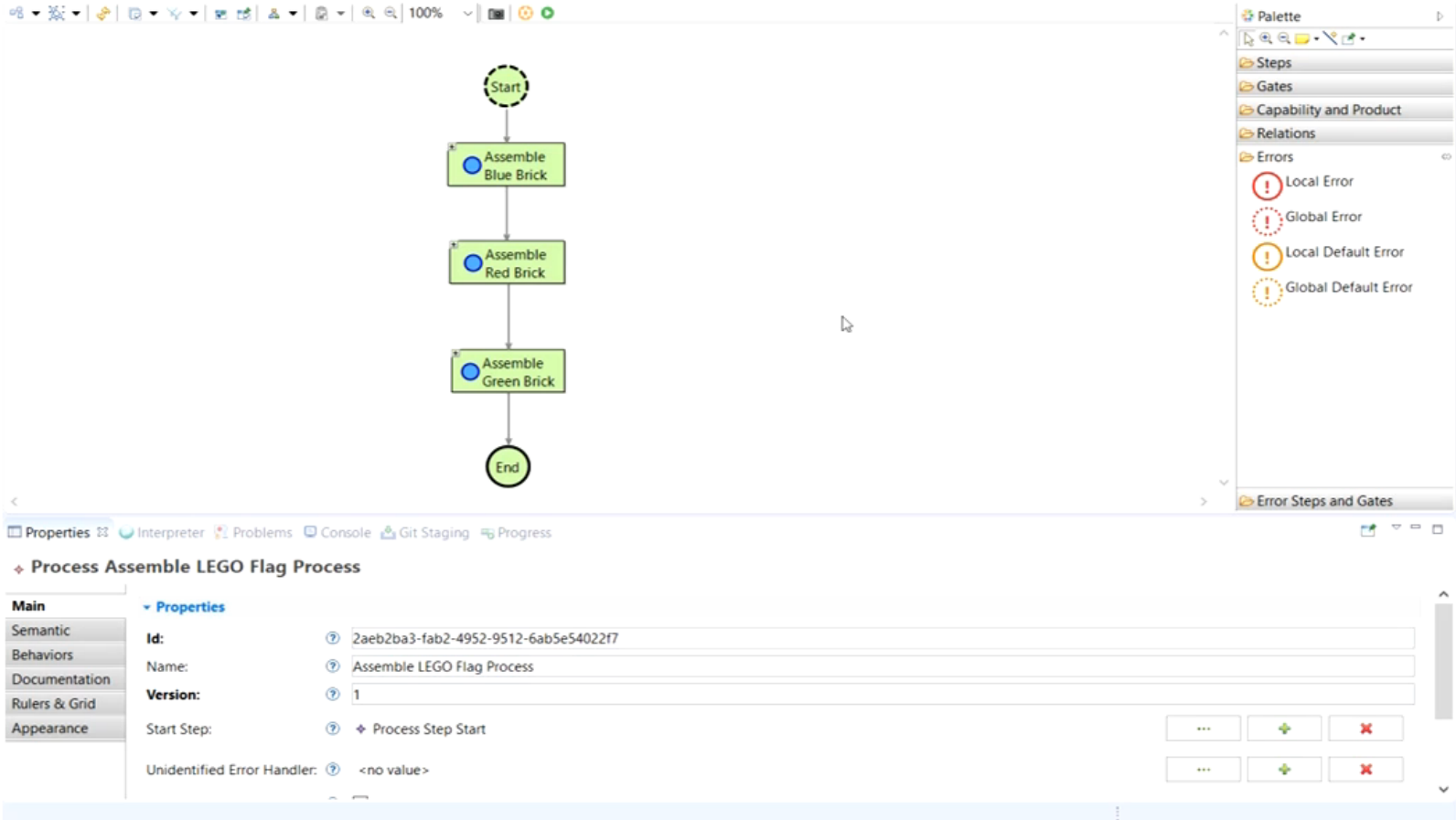
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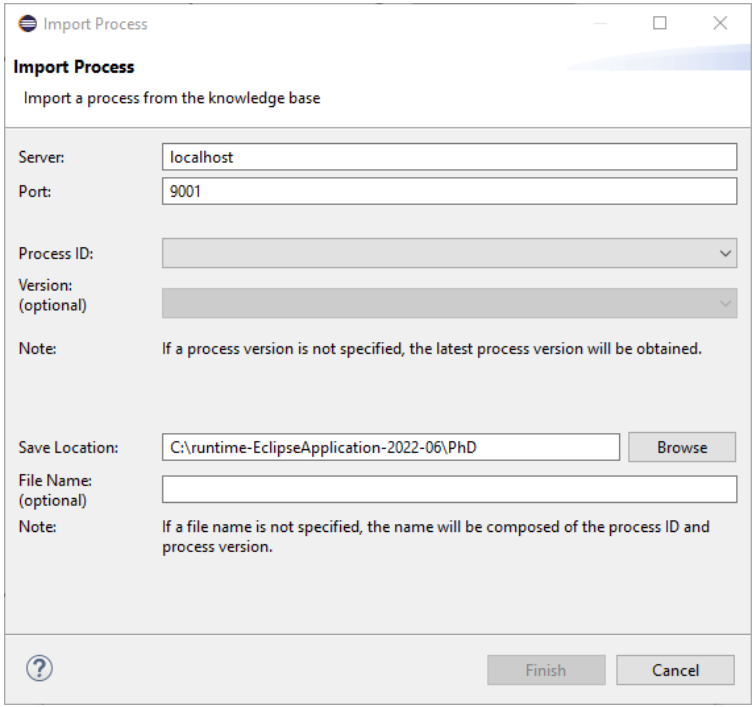
Eclipse Sirius – Additional Functionalities

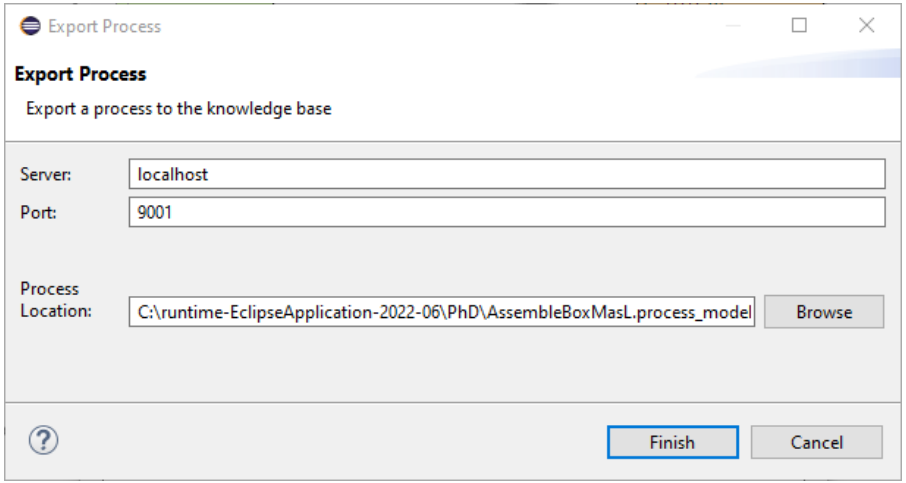
- **Tool Usage**
- **Import/Export**
- **Instruction Generator**
- **Documentation Generator**
- **Capability Repository**
- **Filter Function**
- **Modeling Layers**

MultiProLan – Tool Usage

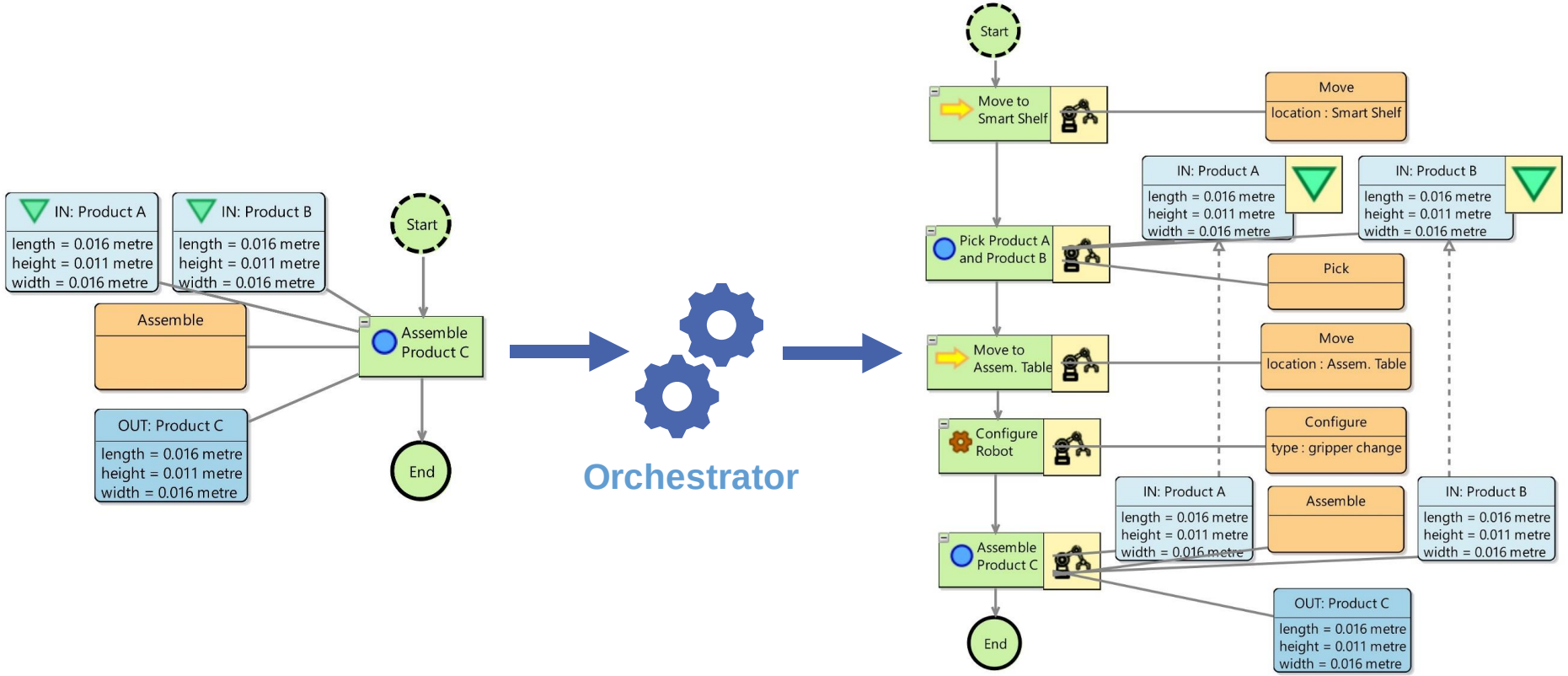


Import/Export Process Models

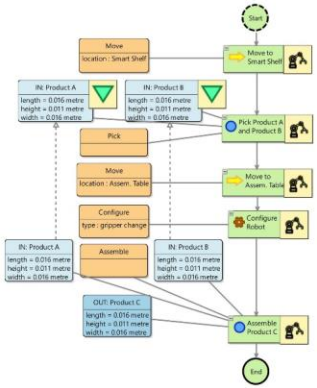


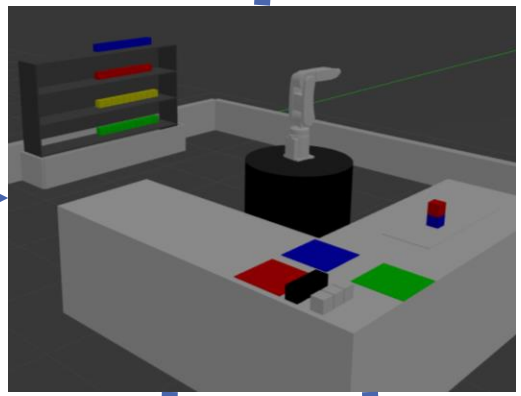


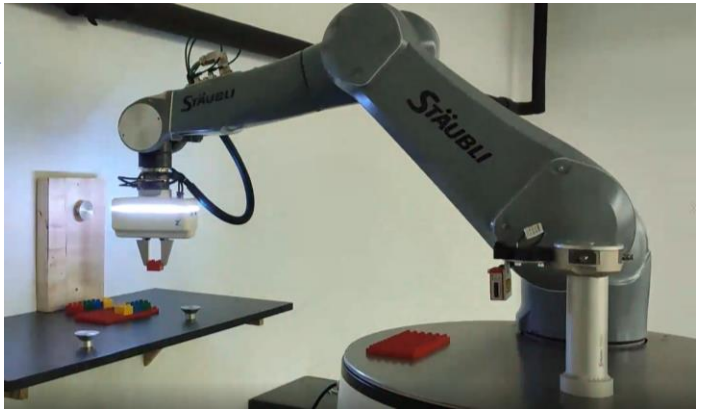
MultiProLan – Models

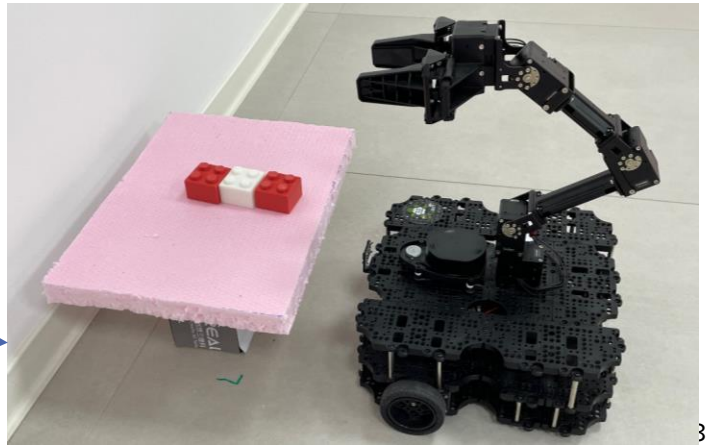


Instruction Generation









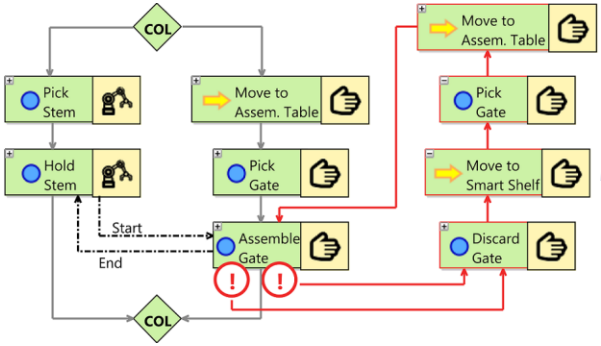




Instruction Generation

```
{
    "orderId": "720f89365d3a",
    "processId": "b6624c68-709d-4afa-a59a-8b0178af8f73",
    "processStepId": "2177e6b8-a653-4ebe-9cf2-fcf55d5a2b65",
    "name": "Pick right side",
    "description": "Pick a frame's right-side wooden plank.",
    "image": "<path>/PickRightSide.jpg",
    "video": "<path>/PickRightSide.avi",
    "capability": {
        "id": "e1a68dce-698c-41fc-af0d-20ff9e1ac093",
        "name": "Pick",
        "requiresStorage": true,
        "parameters": []
    },
    "inputProducts": [{
        "id": "31b93c3f-7ac1-4116-b1f5-b69a730f9b24",
        "name": "Right side",
        "quantity": 1,
        "isStored": true,
        "storage": {
```

Production Documentation Generation



Potential
Failure Mode and Effect Analysis
Process FMEA

FMEA number:_____
Page: 1 **of** 1

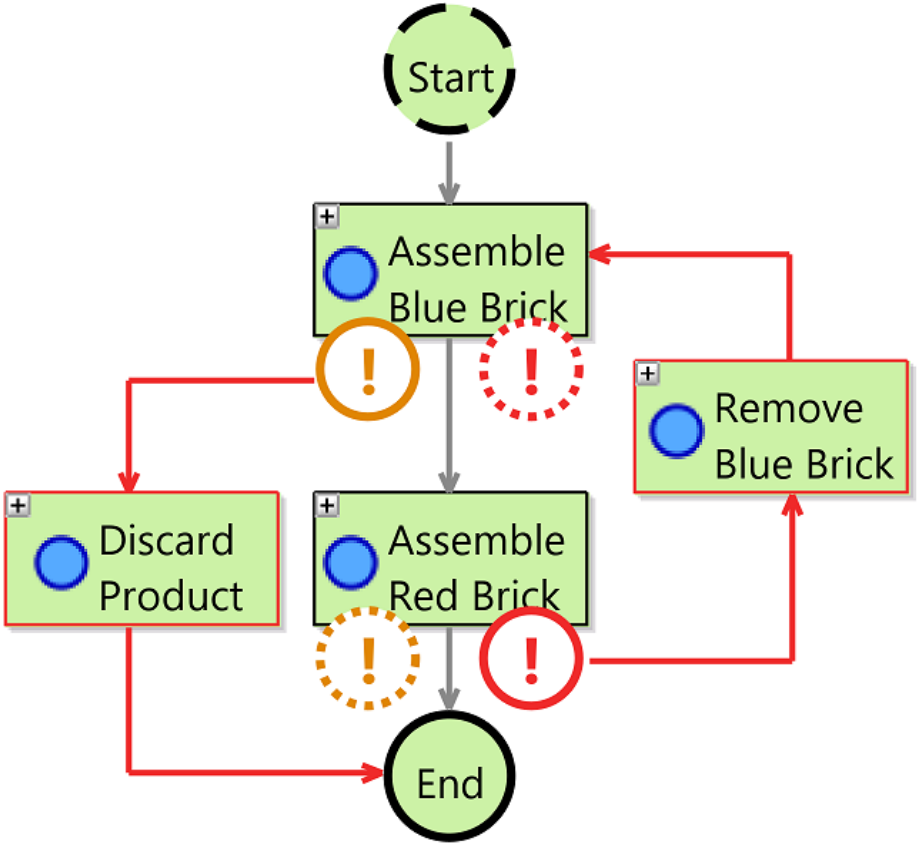
Item: Assemble of stem and gate
Model Year(s)/Programs(s):_____
Core Team: Author A, Author B, Author C

Process Responsibility:_____
Key Date: 2.1.2021.

Prepared By:_____
FMEA Date: (Orig.)_____ **(Rev.)**_____

Process Function / Requirements	Potential Failure Mode	Potential Effect(s) of Failure	Severity	Classification	Potential Cause(s) / Mechanism(s) of Failure	Occurence	Current Control		Detection	R.P.N.	Recommended Action(s)	Responsibility and Target Completion Date	Action Results				
							Prevention	Detection					Actions Taken	Severity	Occurence	Detection	R.P.N.
Assemble Gate	Gate thread has been cracked	Water could leak during the gate valve usage	9	A	Excessive use of force	2	Worker's training	Add additional preasure inspection	4	72							
Assemble Gate	Gate thread has been badly made	Stem and gate cannot be assembled	10	B	Badly made metal	2	Add additional tests after gates are made	-	1	20							

Multi-Layer


Start
Assemble Blue Brick
Remove Blue Brick
Discard Product
Assemble Red Brick
End

Benefits of EMF usage and radip tool prototyping

- A **few days** for the first tool prototype
- Decreases the **cost** for the tool prototype development
- Can be **integrated** with a system currently being used
- Users **learn fast** as a DSL is utilized
- **Customizable?**

Thank You!

Any questions?